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PORTFOLIO

Artificial
Intelligence

Renad Muhammad

ABOUT ME

Hi, I’m Renad – a Machine Learning Engineer specialized in Artificial Intelligence and Computer Vision.
I’m passionate about building intelligent systems that solve real-world problems.
I have experience working with deep learning models, fine-tuning pre-trained architectures, and developing applications in areas like emotion detection, signature forgery detection, and assistive technology.

SKILLS

- Keras / Numpy
- TensorFlow / PyTorch
- Computer Vision
- Deep Learning & Machine Learning
- Natural Language Process (NLP)
- QLoRA Fine-tuning
- Git & GitHub

PROJECTS

👓 SMART GLASSES FOR THE VISUALLY IMPAIRED GRADUATION PROJECT

AS MY GRADUATION PROJECT, I DEVELOPED A WEARABLE ASSISTIVE TECHNOLOGY DESIGNED TO ENHANCE THE INDEPENDENCE OF VISUALLY IMPAIRED INDIVIDUALS. THE SYSTEM INTEGRATES DEEP LEARNING AND COMPUTER VISION TECHNIQUES TO PROVIDE MULTIPLE REAL-TIME FEATURES, INCLUDING OBJECT DETECTION, FACE RECOGNITION, EMOTION RECOGNITION, CURRENCY IDENTIFICATION, AND TEXT READING. USING MOBILENETV2 AND OCR, THE DEVICE PROCESSES LIVE CAMERA INPUT AND PROVIDES AUDIO FEEDBACK, IMPROVING ACCESSIBILITY AND USABILITY.

TOOLS: PYTHON, TENSORFLOW, OPENCV, OCR, TTS, RASPBERRY PI
SKILLS: COMPUTER VISION, REAL-TIME AI, MODEL INTEGRATION, DEEP LEARNING
GITHUB:LINK

SIGNATURE FORGERY DETECTION PROJECT

THIS PROJECT FOCUSES ON DETECTING FORGED SIGNATURES USING DEEP LEARNING TECHNIQUES TO ENHANCE DOCUMENT SECURITY AND PREVENT FRAUD. A SIAMESE NETWORK WITH TENSORFLOW AND KERAS WAS IMPLEMENTED TO LEARN SIGNATURE SIMILARITY PATTERNS. THE SYSTEM COMPARES GENUINE AND QUESTIONED SIGNATURES TO DETERMINE AUTHENTICITY WITH HIGH ACCURACY.

SKILLS & TOOLS: PYTHON, SIAMESE NETWORK, MOBILENETV2, TENSORFLOW, KERAS, COMPUTER VISION, IMAGE PROCESSING

KEY ACHIEVEMENTS:

- **BUILT AND TRAINED A SIAMESE NEURAL NETWORK FOR SIGNATURE VERIFICATION**
- **PREPROCESSED AND AUGMENTED SIGNATURE IMAGE DATASETS TO IMPROVE MODEL ROBUSTNESS**
- **ACHIEVED HIGH ACCURACY IN DISTINGUISHING GENUINE AND FORGED SIGNATURES**
- **CONTRIBUTED TO ENHANCING SECURITY AND FRAUD PREVENTION IN SIGNATURE-BASED AUTHENTICATION SYSTEMS**
- **GITHUB : LINK**

NATURAL LANGUAGE PROCESSING PROJECT

NLP COMBINES COMPUTATIONAL LINGUISTICS AND MACHINE LEARNING TO ENABLE COMPUTERS TO UNDERSTAND, INTERPRET, AND RESPOND TO HUMAN LANGUAGE. THIS COLLECTION OF PROJECTS FOCUSES ON DEVELOPING MODELS AND ALGORITHMS THAT PROCESS TEXT DATA, PERFORM TASKS SUCH AS SENTIMENT ANALYSIS, TEXT CLASSIFICATION, MACHINE TRANSLATION, AND NAMED ENTITY RECOGNITION.

SKILLS & TOOLS: PYTHON, SCIKIT-LEARN, TENSORFLOW, TRANSFORMERS, NLP, TEXT CLASSIFICATION, SENTIMENT ANALYSIS

KEY ACHIEVEMENTS:

- **BUILT SENTIMENT ANALYSIS MODELS WITH HIGH ACCURACY**
- **DEVELOPED MACHINE TRANSLATION PIPELINES**
- **FINE-TUNED PRE-TRAINED MODELS FOR NER**
- **GITHUB : LINK**

BREAST CANCER PREDICTION PROJECT

AS PART OF MY MACHINE LEARNING WORK, I DEVELOPED A PREDICTIVE MODEL TO CLASSIFY BREAST TUMORS AS BENIGN OR MALIGNANT USING THE WISCONSIN DIAGNOSTIC BREAST CANCER DATASET. I PERFORMED DATA PREPROCESSING AND FEATURE SELECTION, THEN TRAINED AND COMPARED MULTIPLE CLASSIFIERS INCLUDING LOGISTIC REGRESSION, DECISION TREES, AND SUPPORT VECTOR MACHINES. THE MODEL WAS EVALUATED USING ACCURACY, PRECISION, AND RECALL, DEMONSTRATING ITS STRONG DIAGNOSTIC POTENTIAL.

TOOLS: PYTHON, SCIKIT-LEARN, PANDAS, MATPLOTLIB, SEABORN
SKILLS: DATA PREPROCESSING, FEATURE ENGINEERING, MODEL SELECTION, EVALUATION METRICS
GITHUB : LINK

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